

Multi-Stage Multi-Hop Retrieval

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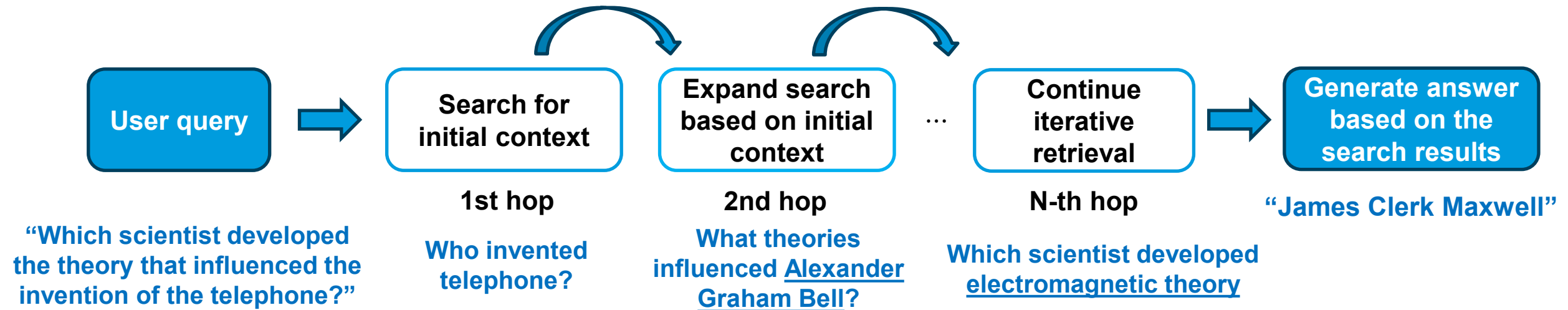
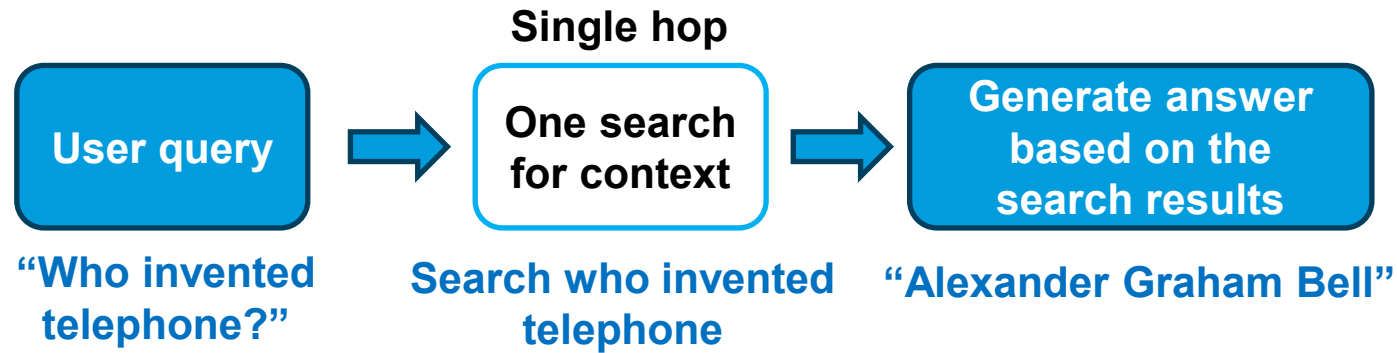
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Outline

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Multi-Hop Retrieval Example (with Query Reformulation)



Research Gaps in Multi-Hop Retrieval

- **Static Query Embeddings:** most existing retrieval models use fixed query embeddings, difficult to adapt to new information.
- **Retriever Staticness:** Many dense retrievers (e.g., DPR, GTR) are used without further fine-tuning, making them poorly suited for multi-hop tasks that depend on subtle reasoning across steps.
- **Error Propagation:** If first-hop retrieval fails, all subsequent retrieval steps fail.
- **Rigid Hop Control:** Most retrieval systems use a fixed number of hops or top-k documents per step. Confidence-based adaptive stopping or fallback mechanisms remain uncommon and underutilized.
- **Efficiency Bottleneck in Query Reformulation:** Text-based query reformulation introduces additional generation overhead and may break semantic continuity.
- **Solution → Retriever Iterative Training + Context-Aware Embedding Refinement**, to enable dynamic query evolution based on retrieval feedback, with fallback mechanisms for low-confidence cases, while being computationally efficient during the refining stage.



Methodology (Retriever Iterative Training)

First stage: Retriever Iterative Training

- **DPR encoders** (Question/Context Encoders)
- **Triplet:** (query, positive, negative)
- Data generated using ground truths (positives) and random negatives (easy-negatives).
- For each training round:
 - Re-encode with the trained encoder from last iteration (or the pretrained encoder).
 - Train with contrastive loss to improve separation between positive and negative documents in embedding space.
- Repeat for a predefined number of iterations to progressively refine the encoder with increasingly challenging negatives.

Methodology (Context-Aware Multi-Hop Retrieval)

Second stage: Context-Aware Multi-Hop Retrieval with Attention Refinement

- **FAISS index:** nearest neighbor search using inner product similarity
- Use fine-tuned **Question Encoder** and **FAISS** to find **top-k** documents
- Refine the query embedding iteratively based on the retrieved context using a lightweight **attention + MLP mechanism**:

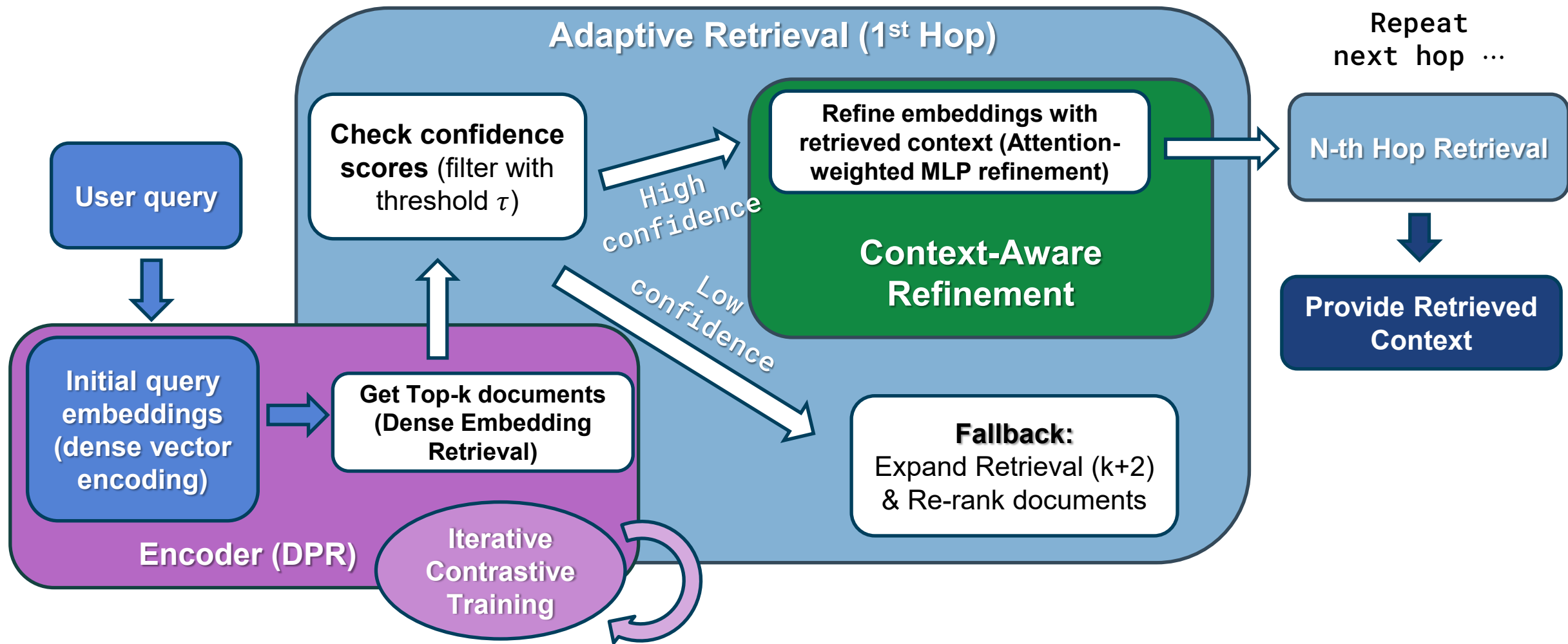
$$Q_{\text{refined}} = \alpha \cdot Q_{\text{original}} + (1 - \alpha) \cdot \text{MLP}(\text{WeightedSum}(R_1, R_2, \dots, R_k)),$$

where α is a weighting factor controlling the strength of the refinement.

- **High-confidence:** Attention-weighted refinement mechanism to refine query
- **Low-confidence:** Fallback expands retrieval (k+2) and re-ranks to prevent semantic drift.



Iterative Encoder Training and Adaptive Query Refinement



Experiment Setup

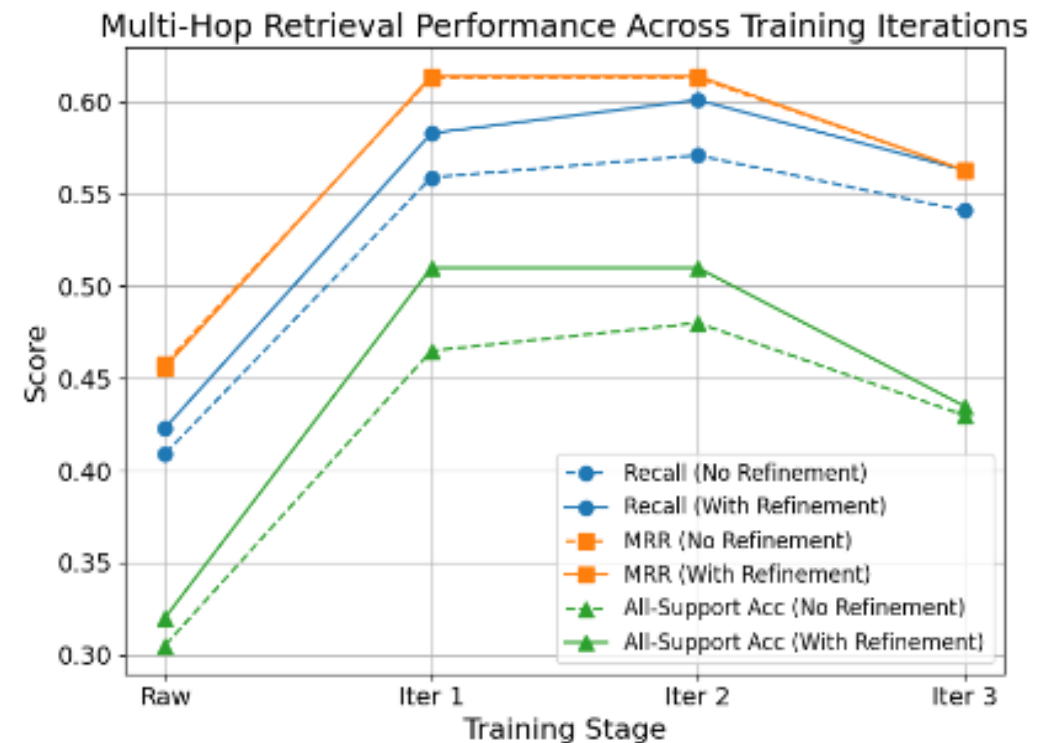
- Single GPU (Colab), CPU: FAISS indexing and retrieval
- Embedding dimension: 768
- **Dataset:** HotpotQA (Full Wiki), 2WikiMultihopQA
 - **Question:** user query
 - **Context** paragraphs: reference documents
 - **Supporting facts:** annotated sentences required for reasoning (ground truth).
 - **Answer:** correct final response, available but not used in our framework since we do not focus on answer generation.
 - Data Split: 800 training samples, 200 test samples



Overall performance (HotpotQA)

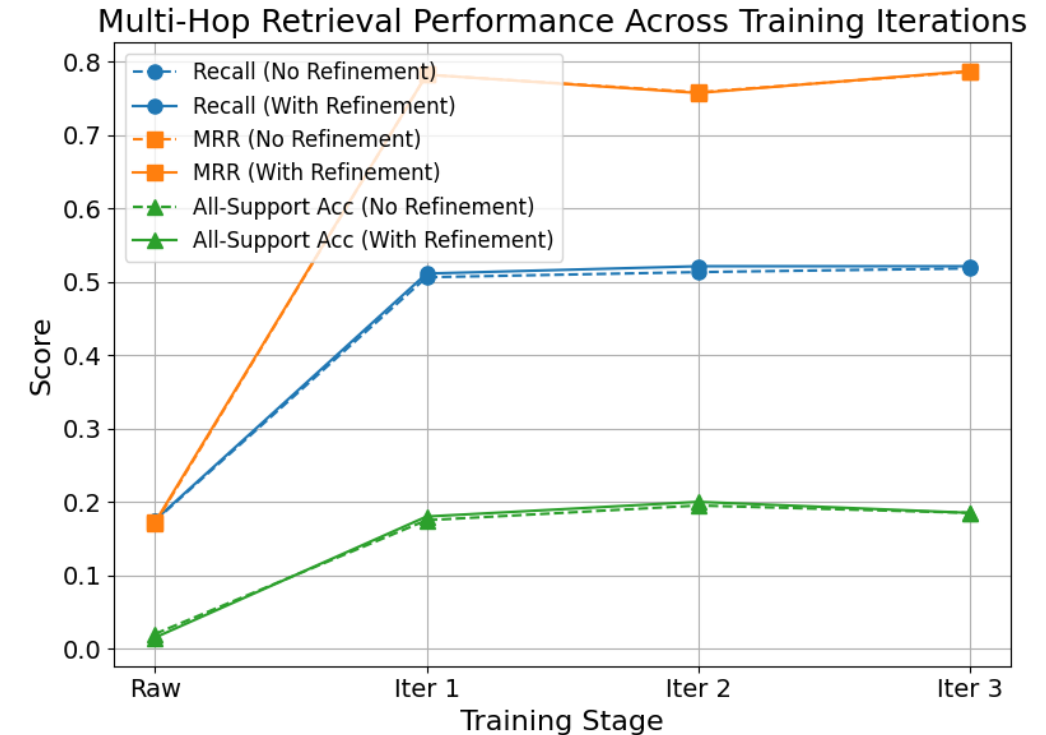
- **Confidence Score**
 - Average similarity over Top-5, num_hops starts with 2
 - Threshold $\tau = 0.6$
- **Evaluation Metrics:**
 - Recall, MRR, All-Support Accuracy

Setup	Avg Recall (%)	MRR (%)	All-Support Acc (%)
Raw encoder, no refinement ($\alpha = 1$)	40.9	45.8	30.5
Raw encoder, with refinement ($\alpha = 0.7$)	42.3	45.6	32.0
Trained (iter=1), no refinement	55.9	61.3	46.5
Trained (iter=1), with refinement	58.3	61.4	51.0
Trained (iter=2), no refinement	57.1	61.3	48.0
Trained (iter=2), with refinement	60.1	61.4	51.0
Trained (iter=3), no refinement	54.1	56.3	43.0
Trained (iter=3), with refinement	56.3	56.3	43.5



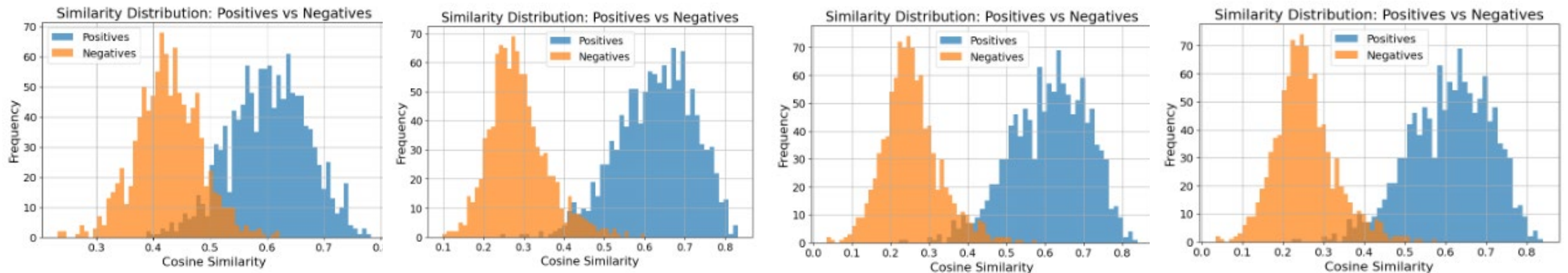
Overall performance (2WikiiMultihopQA)

Setup	Avg Recall (%)	MRR (%)	All-Support Acc (%)
Raw encoder, no refinement ($\alpha = 1$)	17.4	17.0	2.0
Raw encoder, with refinement ($\alpha = 0.7$)	17.5	17.2	1.5
Trained (iter=1), no refinement	50.6	78.2	17.5
Trained (iter=1), with refinement	51.1	78.2	18.0
Trained (iter=2), no refinement	51.3	75.8	19.5
Trained (iter=2), with refinement	52.1	75.7	20.0
Trained (iter=3), no refinement	51.8	78.6	18.5
Trained (iter=3), with refinement	52.1	78.7	18.5



Impact analysis on Iterative training

- Cosine similarity distributions of positive and negative samples.
- The **effects of contrastive learning on DPR**: Improved separation on the embedding space observed over 0 to 3 iterations of training.



Success Case Analysis

Success

```
✓ High confidence at hop 1, score: 0.5738 (avg: 0.5738, margin: 0.0555)
Cosine drift from Q0 at hop 1: 0.6373
Refined query FAISS score: 0.4613
⚠ Low confidence at hop 2, score: 0.3489 (avg: 0.3489, margin: 0.0006)
Q: Katherine Jackson is the matriarch of what American family of signers who originated in Gary, Indiana?
Ground truth sentences:
['Katherine Esther Jackson (born Kattie B. Screws; May 4, 1930) is the matriarch of the Jackson family.', 'The Jackson
Hop 1:
Top 1: Katherine Jackson - Katherine Esther Jackson (born Kattie B. Screws; May 4, 1930) is the matriarch o...
Top 2: Elizabeth Christ Trump - Elizabeth Christ Trump (born Elisabeth Christ; October 10, 1880 - June 6, 1966) ...
Top 3: Mary Bonauto - Mary L. Bonauto (born June 8, 1961) is an American lawyer and civil rights advoc...
Top 4: Katherine Anne Porter - Katherine Anne Porter (May 15, 1890 - September 18, 1980) was an American journa...
Top 5: E. Millicent Sowerby - Emily Millicent Sowerby (September 7, 1883 - October 23, 1977) was a bibliograph...
Hop 2:
Top 1: John Gardner Ford - Married to Juliann Felando Ford on April 29, 1989 and they have two sons, Chris...
Top 2: Gerald Rudolff Ford - Gerald Rudolff Ford (December 9, 1890 - January 26, 1962) was a Grand Rapids bus...
Top 3: Debra Opri - Other high-profile clients have included Michael Jackson's parents, Joseph Jack...
Top 4: Jonathan Daniels - Jonathan Myrick Daniels (March 20, 1939 - August 20, 1965) was an Episcopal semi...
Top 5: Jackson family - The Jackson family is an American family of singers who originated in Gary, Indi...
Hop 3:
Top 1: Aroline Sanborn - She was the wife of Dr. Samuel Bartlett (1811-1865), the great grandson of Josi...
Top 2: Kenneth and Mamie Clark - Kenneth Bancroft Clark (July 14, 1914 - May 1, 2005) and Mamie Phipps Clark (Apr...
Top 3: Jill Biden - Jill Tracy Biden (née Jacobs , previously Stevenson; born June 3, 1951) is an Am...
Top 4: Judy Nelson - Judy Hill Nelson is an American author best known for her 1983-1991 romance (whi...
Top 5: John Gardner Ford - John "Jack" Gardner Ford (born March 16, 1952) is the second child and second so...
```

```
--- Evaluation Results for Sample 3 ---
Recall@15: 1.00
Precision@15: 0.13
MRR: 1.00
All Support Accuracy (titles): True
Stepwise Accuracy per hop: [True, True]
Average Stepwise Accuracy: 1.00
```

Success

```
✓ High confidence at hop 1, score: 0.7213 (avg: 0.7213, margin: 0.0156)
Cosine drift from Q0 at hop 1: 0.9650
Refined query FAISS score: 0.6767
✓ High confidence at hop 2, score: 0.6590 (avg: 0.6590, margin: 0.0010)
Cosine drift from Q0 at hop 2: 0.9302
Refined query FAISS score: 0.6472
Q: Were Jolt Cola and Diet Mountain Dew both sugar-free drinks, as advertised?
Ground truth sentences:
['It was targeted towards students and young professionals, stressing its use as a stimulant in a similar manner as energy drin
Hop 1:
Top 1: Jolt Cola - Jolt Cola was a carbonated soft drink produced by The Jolt Company, Inc. (later ...
Top 2: Pepsi-Cola Made with Real Sugar - The Pepsi-Cola Made with Real Sugar, formerly called Throwback, is a brand of so...
Top 3: Mountain Dew - Mountain Dew (sometimes stylized as Mtn Dew) is a carbonated soft drink brand pr...
Top 4: Pepsi-Cola Made with Real Sugar - The drinks, called Pepsi Throwback and Mountain Dew Throwback, are named as suc...
Top 5: Diet Pepsi - Diet Pepsi and Diet Pepsi Classic Formula Blend (stylized as diet PEPSI CLASSIC ...
Hop 2:
Top 1: Diet Mountain Dew - It was formerly known as "Sugar-Free Mountain Dew" until 1986, when it was give...
Top 2: Pepsi-Cola Made with Real Sugar - As of June 2014, Pepsi Throwback has been replaced in some areas of the United ...
Top 3: Diet Mountain Dew - Diet Mountain Dew is a no-calorie Mountain Dew that was first introduced in 1986...
Top 4: Mello Yello - Mello Yello is a highly-caffeinated, citrus-flavored soft drink produced and dis...
Top 5: Surge (drink) - Surge (sometimes styled as SURGE) is a citrus flavored soft drink first produced...
```

```
--- Evaluation Results for Sample 5 ---
Recall@10: 0.50
Precision@10: 0.10
MRR: 0.17
All Support Accuracy (titles): True
Stepwise Accuracy per hop: [False, True]
Average Stepwise Accuracy: 0.50
```



Failure Case Analysis

Failure

```
✓ High confidence at hop 1, score: 0.6871 (avg: 0.6871, margin: 0.0377)
Cosine drift from Q0 at hop 1: 0.7792
Refined query FAISS score: 0.5891
⚠ Low confidence at hop 2, score: 0.5166 (avg: 0.5166, margin: 0.0029)
Q: The eastern terminus of New York State Route 67 is in a town with what population as of the 2010 census?
Ground truth sentences:
[' Its eastern terminus is at the Vermont state line in Hoosick, where it continues eastward as Vermont Route 67 (VT 67).']

Hop 1:
Top 1: New York State Route 67 - New York State Route 67 (NY 67) is an east-west state highway in eastern New Yor...
Top 2: New Jersey Route 67 - Route 67 is a short state highway entirely within the community of Fort Lee in B...
Top 3: New York State Route 67 - The western terminus of the route is at an intersection with NY 5 in the town o...
Top 4: New York State Route 221 - The eastern terminus is at a junction with NY 26 and NY 41 in the town of Wille...
Top 5: New York State Route 221 - New York State Route 221 (NY 221) is a state highway in Cortland County, New Yor...

Hop 2:
Top 1: New York State Route 67 - Its eastern terminus is at the Vermont state line in Hoosick, where it continue...
Top 2: Northern State Parkway - The eastern terminus is at New York State Route 347 (NY 347) and NY 454 in Haup...
Top 3: New York State Route 241 - The southern terminus of the route is at an intersection with NY 394 in the ham...
Top 4: New York State Route 221 - Its western terminus is at an intersection with NY 38 in the town of Harford....
Top 5: New York State Route 221 - Near its western end, NY 221 intersects New York State Route 200....

Hop 3:
Top 1: New York State Route 241 - New York State Route 241 (NY 241) is a north-south state highway in Cattaraugus ...
Top 2: New York State Route 285 - New York State Route 285 (NY 285) was a state highway in Oneida County, New York...
Top 3: U.S. Route 9 in New Jersey - U.S. Route 9 (US 9) is a U.S. highway in the northeast United States, running fr...
Top 4: Yonkers Avenue - The western terminus of the street is at Nepperhan Avenue, which connects to U...
Top 5: New York State Route 285 - The southern terminus of the 6.87 mi route was at an intersection with NY 69 we...
```

```
--- Evaluation Results for Sample 1 ---
Recall@15: 0.50
Precision@15: 0.07
MRR: 0.17
All Support Accuracy (titles): False
Stepwise Accuracy per hop: [False, False]
Average Stepwise Accuracy: 0.00
```



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Failure

```
✓ High confidence at hop 1, score: 0.6856 (avg: 0.6856, margin: 0.0577)
Cosine drift from Q0 at hop 1: 0.6694
Refined query FAISS score: 0.5511
⚠ Low confidence at hop 2, score: 0.4388 (avg: 0.4388, margin: 0.0110)
Q: What was the final Marilyn Manson album to feature the member who took a stage name from a serial killer?
Ground truth sentences:
['The Golden Age of Grotesque is the fifth studio album by American rock band Marilyn Manson.', " It was also their final

Hop 1:
Top 1: Marilyn Manson (band) - Their lineup has changed between many of their album releases; the current memb...
Top 2: Velocity Girl (song) - Shortly preceding its release, Primal Scream's frontman, Bobby Gillespie, left ...
Top 3: Marilyn Manson (band) - Marilyn Manson is an American rock band formed by singer Marilyn Manson and guit...
Top 4: Vile (album) - "Vile" also marks the last album featuring guitarist Rob Barrett until "Kill", ...
Top 5: Marilyn Manson - He is known for his controversial stage personality and image as the lead singe...

Hop 2:
Top 1: QR III - It is the last album to feature lead singer Kevin DuBrow until the 1993 album "...
Top 2: Twiggy Ramirez - Jeordie Osbourne White (born June 20, 1971), once known professionally as Twiggy...
Top 3: Disarm the Descent - Disarm the Descent is the sixth studio album by American metalcore band Killswit...
Top 4: Rehab (Quiet Riot album) - This is the last Quiet Riot studio album to feature lead singer Kevin DuBrow be...
Top 5: Marilyn Manson - Brian Hugh Warner (born January 5, 1969), known professionally as Marilyn Manson...

Hop 3:
Top 1: The Golden Age of Grotesque - It was also their final studio album to feature keyboardist Madonna Wayne Gacy ...
Top 2: Paul Mazurkiewicz - The trio joined Alex Webster and Jack Owen in 1988, forming Cannibal Corpse....
Top 3: Metallica discography - The following is the discography of Metallica, a Los Angeles-based metal band fo...
Top 4: The White Stripes - The band recorded two more albums, "Get Behind Me Satan" in 2005 and "Icky Thum...
Top 5: Quiet Riot 10 - It is their first studio album since 1988's "QR" not to feature longtime and fo...
```

```
--- Evaluation Results for Sample 5 ---
Recall@15: 0.25
Precision@15: 0.07
MRR: 0.09
All Support Accuracy (titles): False
Stepwise Accuracy per hop: [False, False, False, False]
Average Stepwise Accuracy: 0.00
```

Conclusion

- Iterative contrastive learning + embedding refinement reduces error propagation in multi-hop retrieval.
- Achieved ~20% gains in recall and all-support accuracy after 2 training iterations.
- Successful cases show early topic anchoring and semantic consistency.
- Failures reveal limits in shifting between entity types.

Future work

- Compare with advanced baselines (QR-based methods weren't tested due to time limit).
- Improve semantic adaptability to support cross-type reasoning.
- Scale to larger and multi-domain datasets (e.g., more samples, MuSiQue, etc.).
- Refine confidence-based fallback and early exit logic.
- Explore guided refinement using answer-type signals.



Thank you!



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References

- Akari Asai, Kazuma Hashimoto, Hannaneh Hajishirzi, Richard Socher, and Caiming Xiong. Learning to retrieve reasoning paths over wikipedia graph for question answering. arXiv preprint arXiv:1911.10470, 2019.
- Hyunji Lee, Sohee Yang, Hanseok Oh, and Minjoon Seo. Generative multi-hop retrieval. In Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang, editors, Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, pages 1417–1436, Abu Dhabi, United Arab Emirates, December 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.92. URL <https://aclanthology.org/2022.emnlp-main.92/>.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In Proceedings of the 34th Conference on Neural Information Processing Systems (NeurIPS). NeurIPS, 2020.
- Peng Qi, Xiaowen Lin, Leo Mehr, Zijian Wang, and Christopher D. Manning. Answering complex open-domain questions through iterative query generation. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pages 2590–2602, Hong Kong, China, November 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1261. URL <https://aclanthology.org/D19-1261/>.
- lireza Salemi and Hamed Zamani. Evaluating retrieval quality in retrieval-augmented generation, 2024. URL <https://arxiv.org/abs/2404.13781>.
- Zhengliang Shi, Shuo Zhang, Weiwei Sun, Shen Gao, Pengjie Ren, Zhumin Chen, and Zhaochun Ren. Generate-then-ground in retrieval-augmented generation for multi-hop question answering. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 7339–7353, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi:10.18653/v1/2024.acl-long.397. URL <https://aclanthology.org/2024.acl-long.397/>.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In Advances in Neural Information Processing Systems, pages 5998–6008, 2017.
- Weiwen Xu, Huihui Zhang, Deng Cai, and Wai Lam. Dynamic semantic graph construction and reasoning for explainable multi-hop science question answering. In Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli, editors, Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021, pages 1044–1056, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.findings-acl.90. URL <https://aclanthology.org/2021.findings-acl.90/>.
- Yunchang Zhu, Liang Pang, Yanyan Lan, Huawei Shen, and Xueqi Cheng. Adaptive information seeking for open-domain question answering. In Conference on Empirical Methods in Natural Language Processing, 2021. URL <https://api.semanticscholar.org/CorpusID:237502990>.

